

PARTS LIST

Semiconductors:

- IC1 - LM386 low-power audio amplifier
T1 - 2N2219 npn transistor
T2 - BD139 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 2.2-kilo-ohm
R2 - 1-kilo-ohm
R3 - 22-ohm
R4 - 18-kilo-ohm

Capacitors:

- C1, C2, C4, C7 - 100nF ceramic disk
C3 - 470pF ceramic disk
C5 - 220pF ceramic disk
C6, C8 - 220 μ F, 35V electrolytic

Miscellaneous:

- MIC1 - Electrate microphone
CON1 - 2-pin terminal connector
JACK1 - 2-pin connector
XTAL1 - 6.4/7.2MHz crystal
ANT.1 - 10 metres long wire
RFC1, RFC3 - 330 μ H inductor
RFC2 - 2.2mH inductor
- 12V DC regulated power supply/12V battery

can rise to 120mW with 80 percent modulation by using a 10-metre long wire antenna for transmission.

Construction and testing

An actual-size PCB layout for the voice transmitter is shown in Fig. 2

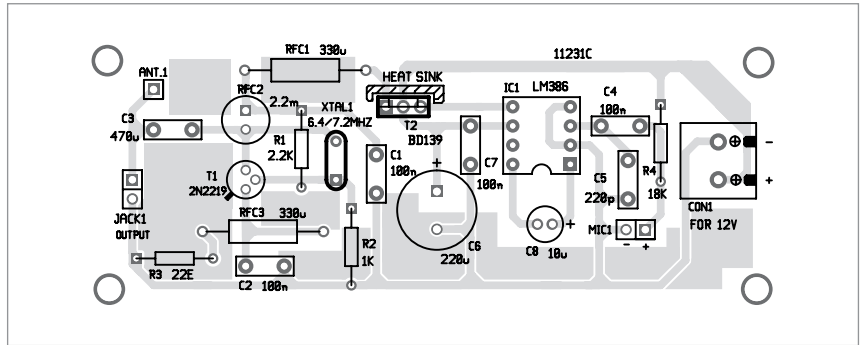


Fig. 3: Components layout of the PCB

and its components layout in Fig. 3. After assembling the circuit on PCB, connect a 12V battery across connector CON1 and take audio output from JACK1. The audio can be heard on a shortwave radio receiver after proper tuning.

Connect MIC1 and speak into it. The same can be listened to on a short-wave radio tuned at 6.4 or 7.2MHz frequency.

You can assemble the simple circuit even on a general-purpose PCB. The RF chokes used are readymade moulded chokes and their values

are not critical. RFC1 and RFC3 can be of any value between 100 μ H and 680 μ H. RFC2 can be 1mH to 4.7mH. A 7.2MHz crystal was used during testing. Use a 12V hum-free regulated power supply.

After constructing the circuit on PCB, house it in a suitable plastic box. **EFY**



Joy Mukherji is an electronics hobbyist who likes designing radio frequency circuits

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